Lab 3.1
FMEA Document and RPN Table

Goal	Build a complete FMEA for the dosing skid and calculate RPN with Copilot in Excel.
You will produce	A ranked FMEA table with RPN, a severity escalation list, an RPN chart and an action list.
Time box	30 minutes
Outcome	LO1 explain FMEA · LO2 automate the document with Copilot in Excel
Case	CW-401 chemical dosing skid

All records are fictional training data for classroom use. They describe no actual facility, person or organisation. Values are teaching assumptions, not approved design criteria. Do not use any output to operate, maintain or approve real equipment.

Before you start

Open the Starter workbook for this lab. Do not open Worked-Reference.xlsx until you have finished your own attempt and checked it. Sign in to Microsoft 365 at https://m365.cloud.microsoft/ with your training account.

The concept in one minute

FMEA asks one question in a disciplined order: what could fail, what happens if it does, why would it happen, and what would catch it first. Each row scores Severity, Occurrence and Detection from 1 to 10, and RPN multiplies the three. Detection is inverted, so a high D means you would probably NOT catch it. RPN ranks work; it never decides acceptability. A Severity 9 row is escalated even when its RPN is the lowest on the sheet, because a safety consequence is not cancelled by being rare or easy to spot.

Step-by-step

Step 1 — Open the workbook and read the record

Open **Starter.xlsx** in this folder. Look at the **FMEA_Input** sheet.

You have four supplied failure modes for the CW-401 dosing skid, each already rated for S, O and D. The RPN and Severity flag columns are deliberately empty. You will fill them.

Open the **Rating_Guide** sheet and read the Detection column carefully. Note that 9-10 means no credible detection before failure. This is the opposite of most scales.

Step 2 — Select the named table

Click any cell inside the FMEA table, then check the Name Box shows you are inside **FMEAInput**. Copilot in Excel works far better against a named table than a loose range.

Step 3 — Open Copilot in Excel and paste the prompt

On the Home ribbon, click **Copilot**. The pane opens on the right.

Open **Lab-3.1-Prompt.pdf** (or the Copilot_Prompt sheet) and paste the whole prompt.

Notice the prompt names the table explicitly and forbids outside knowledge. That single line is what stops Copilot inventing a fifth failure mode.

Step 4 — Read the RPN arithmetic before you accept it

Copilot returns RPN for each row. Do not skim it. Check FM-01 by hand:

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= 7 x 5 x 4 = 140
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If Copilot shows a different number, or shows no arithmetic at all, reject the answer and ask it to show the multiplication for every row.

Step 5 — Insert the RPN chart

Ask Copilot to insert a clustered column chart of RPN by failure mode ID.

Order the bars highest to lowest so the ranking is visible without reading numbers.

Title the chart so a reviewer knows what they are looking at, for example "CW-401 FMEA — RPN by failure mode".

Step 6 — Apply the severity gate yourself

Sort or filter the table on the S column and find every row with S of 9 or higher.

FM-04 has Severity 9 and the lowest RPN of the four. It still escalates.

In the Severity flag column write ESCALATE for that row. If Copilot did not flag it, that is the single most important correction you will make today.

Step 7 — Request actions and evidence gaps

Ask Copilot for one evidence request and one practical follow-up action per row.

Check that no action reads like an approval. "Approve the design" is wrong; "request the calibration certificate for FT-1" is right.

Step 8 — Compare with the worked reference

Only now open **Worked-Reference.xlsx**. Compare your RPN values and your escalation list.

Note where your wording differs. Differences in wording are fine; differences in the numbers or the escalation are not.

Check your own work

Do these by hand before you accept the Copilot output. If a number disagrees, the Copilot answer is wrong until proven otherwise.

- FM-01 RPN by hand: 7 x 5 x 4 = 140.
- FM-02 RPN by hand: 8 x 4 x 6 = 192, the highest of the four.
- FM-04 has the LOWEST RPN (54) but Severity 9, so it must still be escalated.
- No fifth failure mode appeared. Copilot used only the four supplied rows.
- Every supplied S, O and D rating is unchanged.

Engineering boundary

This is a drafting aid, not an approval. The agent must not choose or change a rating, decide that a risk is acceptable, or state that the skid is safe or fit for service. A rating may only be lowered when verified evidence of control effectiveness exists; a planned action is not evidence. Residual ratings stay unassessed until then.

Evidence to retain

Save the final prompt, the workbook, one independent hand check, any correction you made and the name of the human reviewer.

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